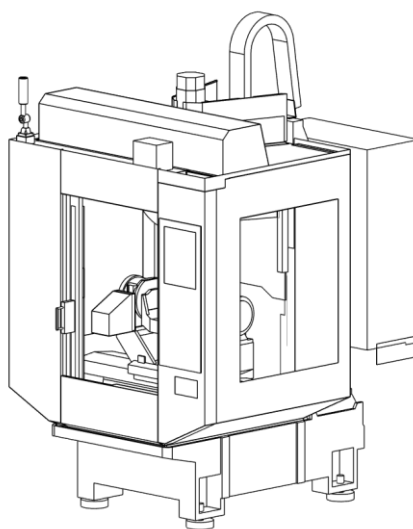




Milling Machine Simulation Setup Guide

Purpose: This document is intended to assist the user in transforming a machine tool model into a version suitable for simulation and collision checking in Fusion. This document is for milling machines – it does not address turning or additive machines

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(Note that Table of Contents is made of hyperlinks. If viewing this document in Word, use ctrl+click to jump to pages in the Table of Contents, and use ctrl+home to return to top of document)

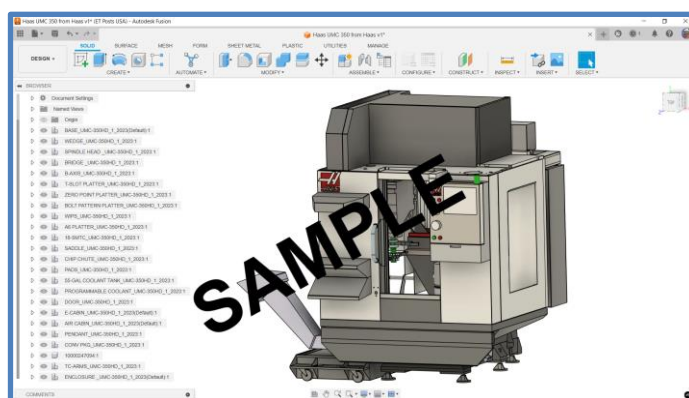
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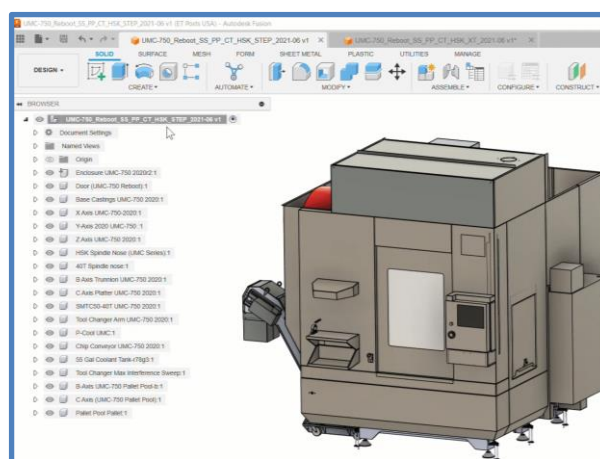
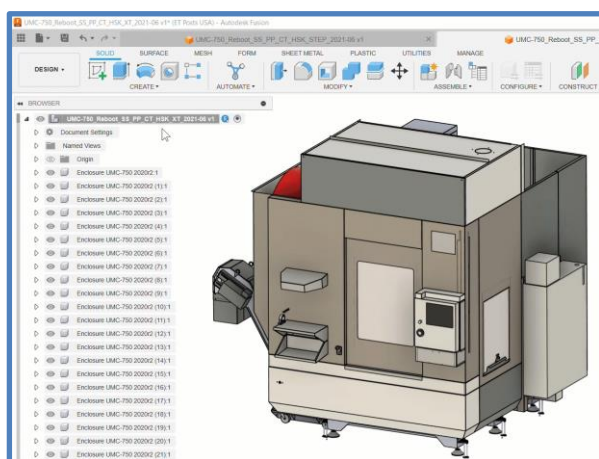
1. IMPORT AND POSITION THE MODEL OF MACHINE TOOL

A. Considerations when importing models

Often you can get machine models from the manufacturer's web site, or by requesting them from the manufacturer. The image below is a machine downloaded directly from the Haas web site.



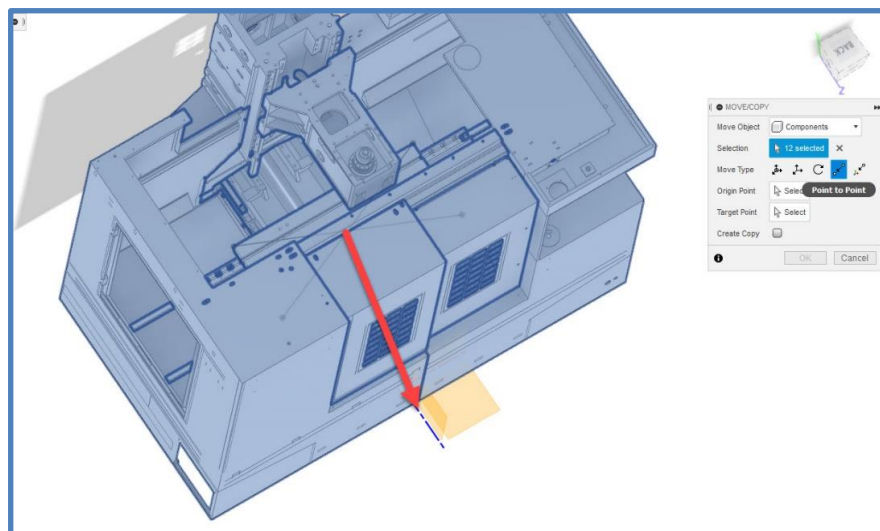
When you have multiple file types available for the same machine, know that some will be easier to work with when performing the simplification described in this document. The image below on the left is a Parasolid file that was 72K, and the one on the right a STEP file that was 34K. The file on the right will be *much* easier to work with because there are many less 'bits' to be organized.



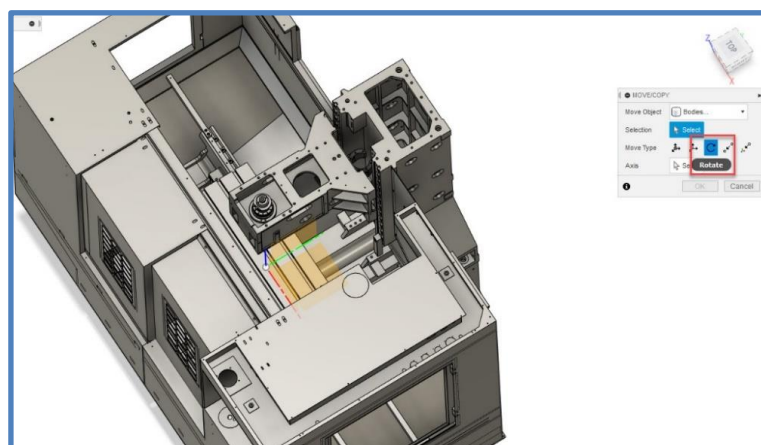
Remember that if the manufacturer supplied the following, it is important to include them:

- Real machine colors
- Manufacturer logo (If it was part of the model supplied by the manufacturer.)
- Controller in the model.

B. Move the entire model to the origin position



C. Rotate the entire model to properly orient it along XYZ origin



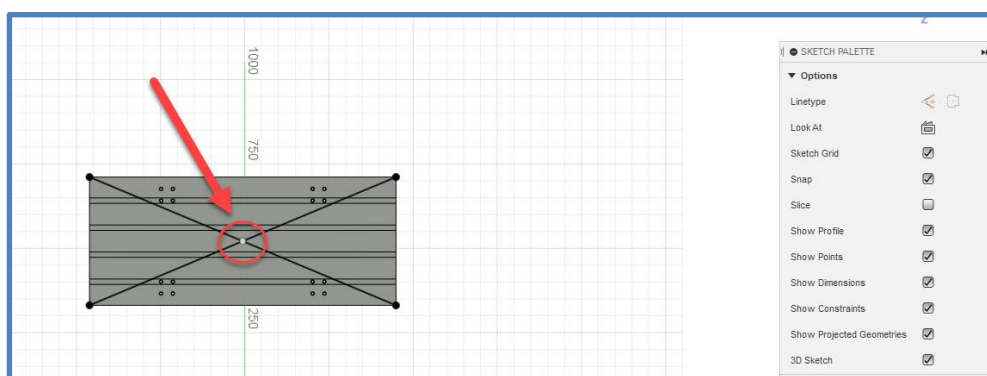
To ensure the model works with machine simulation, the CAD XYZ should align with the physical machine XYZ. This will ensure all machine axes move correctly when loaded into simulation.

TIP: When rotating around an axis, you can expand the 'Origin' item in the Browser and select an axis.

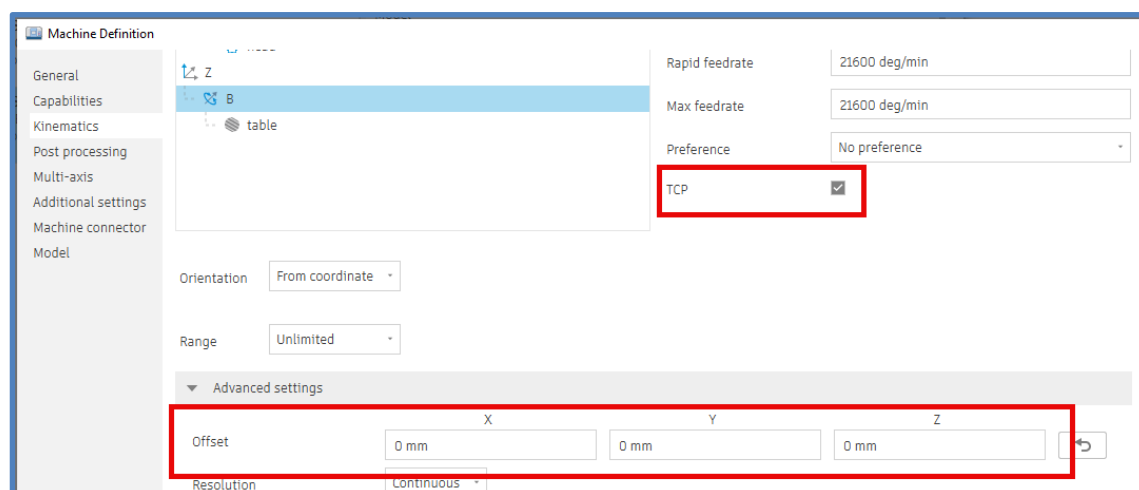
D. Set the Origin for the model

Note that this process will vary depending on the machine configuration – Head/Head, Table/Head, Table/Table – determines where you should position the machine origin.

Commonly you will define a point (=zero position) on the machine table center, then position the model so that point is at coordinates 0, 0, 0.

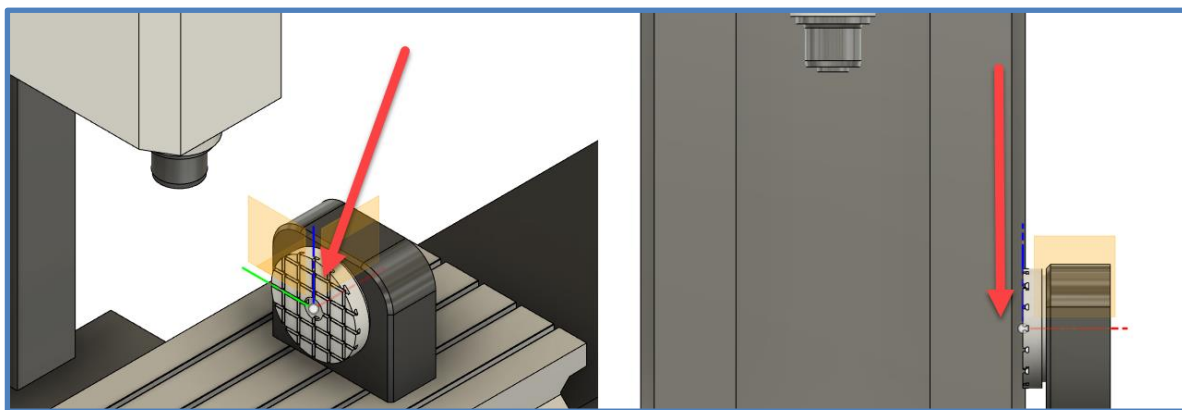


By positioning the origin as shown, when you later setup the Kinematics in the Machine Definition, you can set the 'Offset' value to zero.



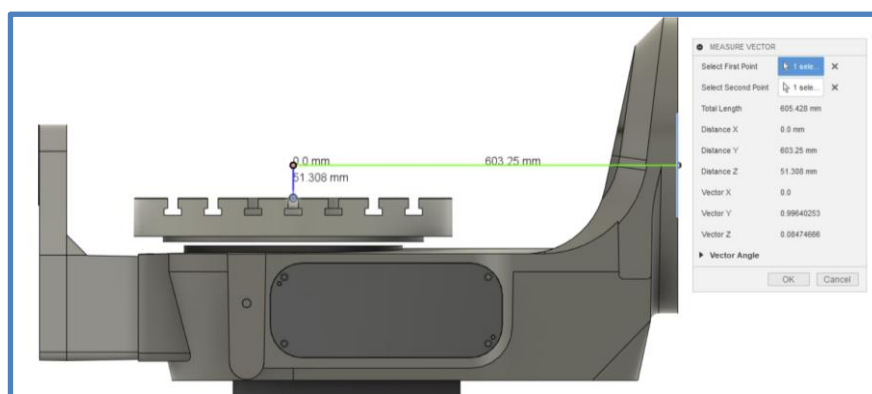
Machine configurations that have the 4th axis rotating around X or Y axis should be positioned so the center of the rotary axis is at the model origin as shown in the picture below.

Note that non-TCP, 4th axis machines, **must** have the origin at the face of where the part will mount – as shown in the picture below.

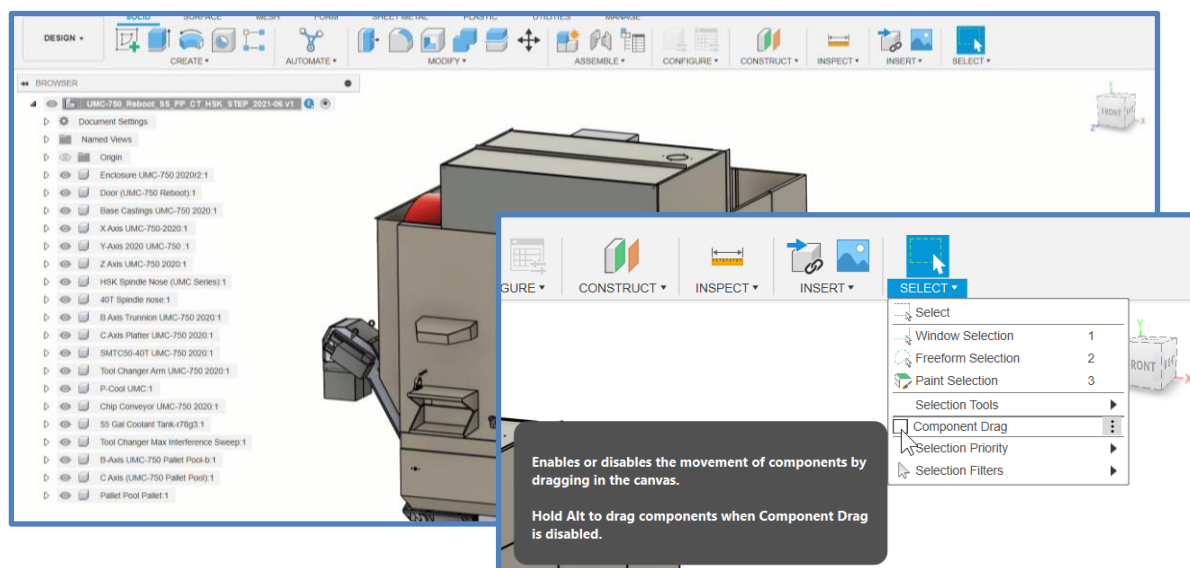


Non-TCP machines that are 4-axis or 5-axis should have the origin at the top center of the table. Again, this is important so that when you later setup the Kinematics in the Machine Definition, you can set the 'Offset' value to zero.

The exception is when there is an offset between the centers of rotation. In the picture below, an offset of 51.308 is shown. For this condition, with non-TCP machines, the origin should be set to the intersection of the rotaries for the 4th and 5th axis, and the offset value (51.308) entered when setting the kinematics. This offset distance is normally only in the Z axis and is sometimes called the 'Pivot Distance'.



Once you have the machine model position as required, it is advised that you 'Ground' the model, to avoid unwanted drags and/or movements. Additionally, it is advised to uncheck the 'Component Drag' setting. Both operations are shown in the pictures below. The picture below shows this being done.



E. Saving the file now is recommended

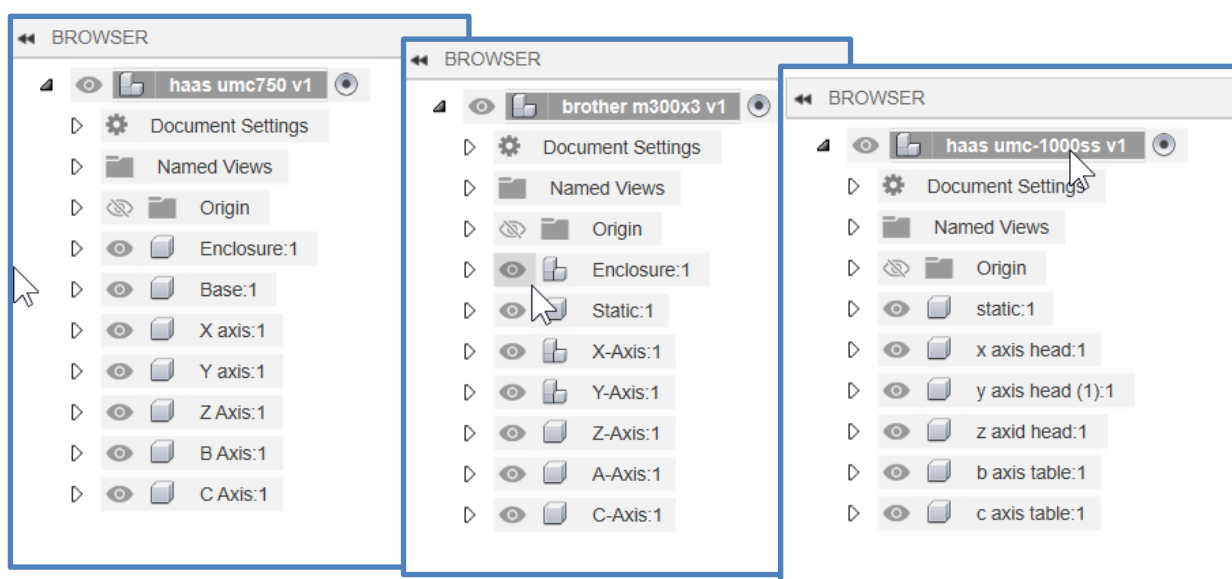
Before continuing, you may want to do a 'Save As' and save the file with a new name. This will give you a convenient place to return to if you have difficulties or errors in the process going forward.

2. GROOM THE MODEL FOR USE BY FUSION

A. Define components and group the machine parts

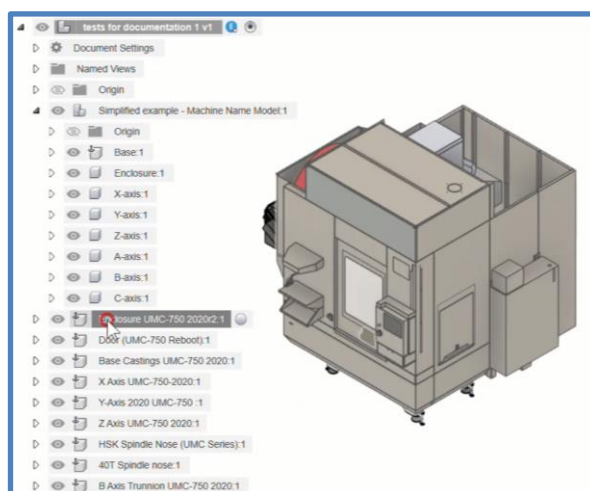
It is necessary to organize, categorize, and clean unnecessary detail from these machine models.

Start by creating a new component with a name that represents the machine and model. Then create components beneath it as shown in the examples below.



Only use names like these: X-axis, Y-axis, Z-axis, A-axis, B-axis, C-axis, Base/Static/Enclosure.

Then drag and drop the components for the various machine parts into the appropriate components. The picture below is an example of how to move the components.



The result should be all the components in the file are inside the newly created components named for machine bits. Later, when using the Machine Builder, this will make the task of linking these components to axis of motion much easier.

There shouldn't be any needs for sketches, so they can be deleted.

There should be no unstitched surfaces/bodies: these will need to be turned into solids or deleted.

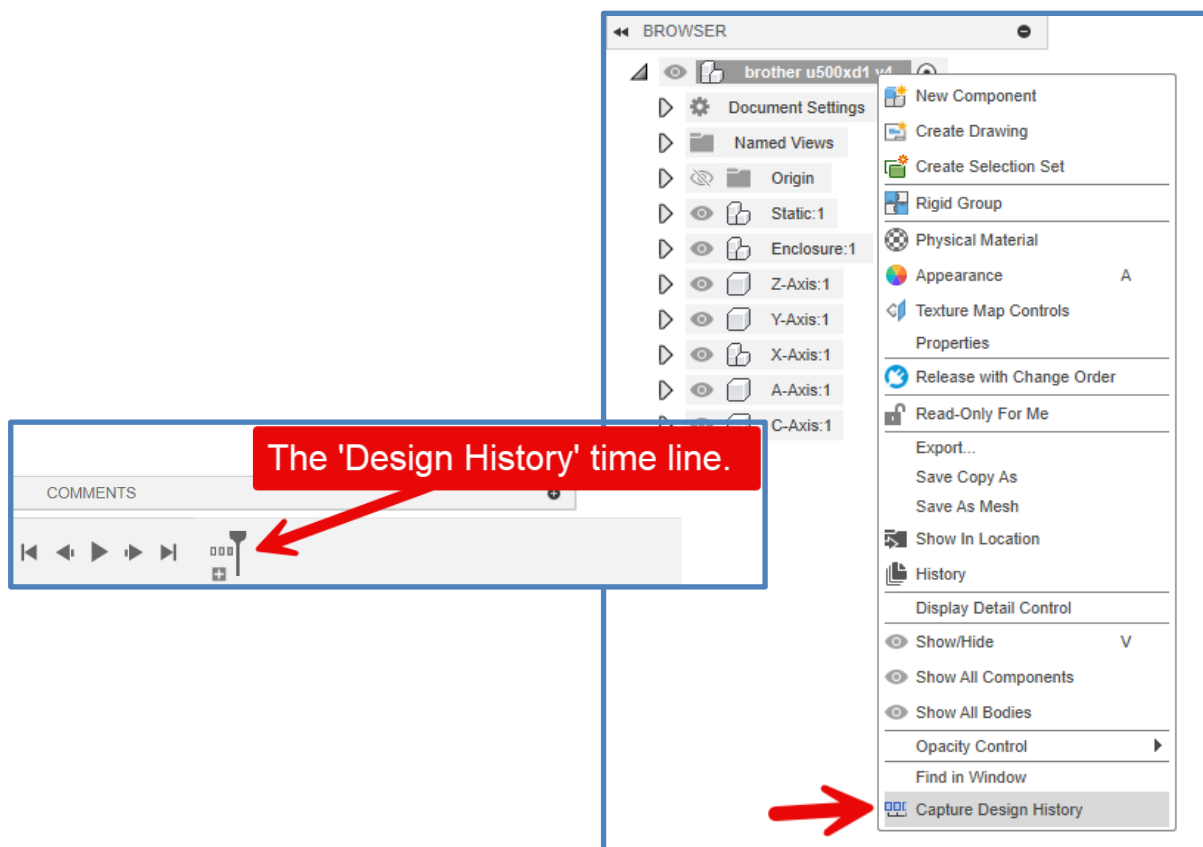
Note that if a component is a surface, Fusion will not check for collisions, so convert it to a body instead of a surface.

It is recommended to turn off all construction objects (planes, origins, axes etc..), sketches and images if not required for simulation. This will ensure the model looks as natural as possible and the view will not be obscured by any design/construction entities.

B. Refinement of the model

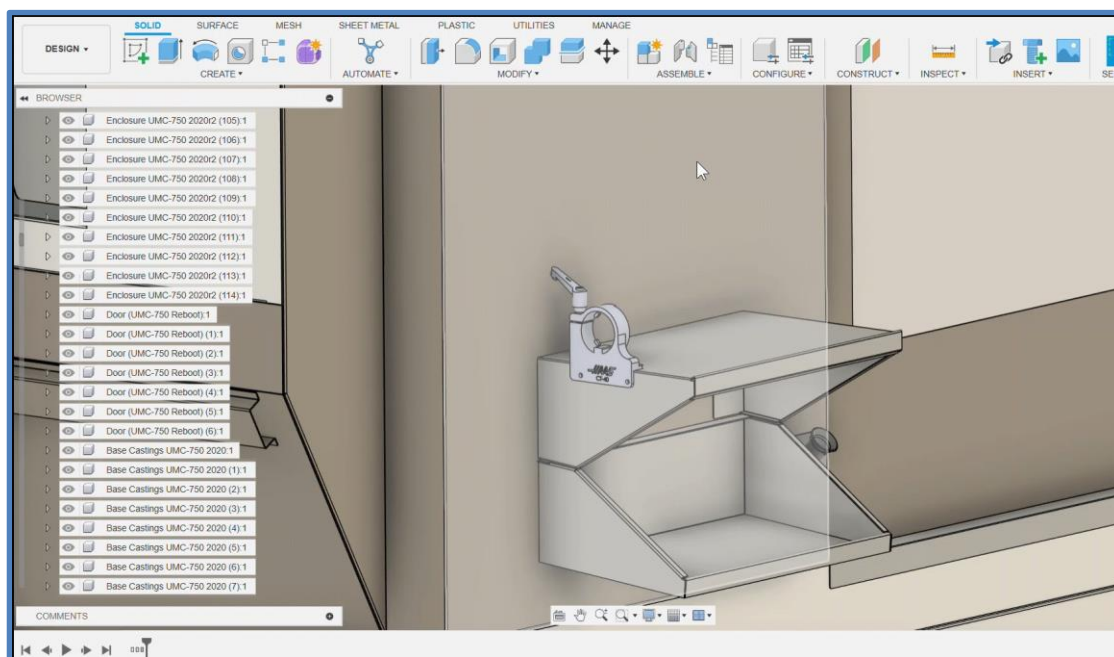
After the model has been oriented, simplified and properly grouped, some further analysis must be performed.

Begin by turning on 'Capture Design History'. This will allow the use of 'Simplify' commands.

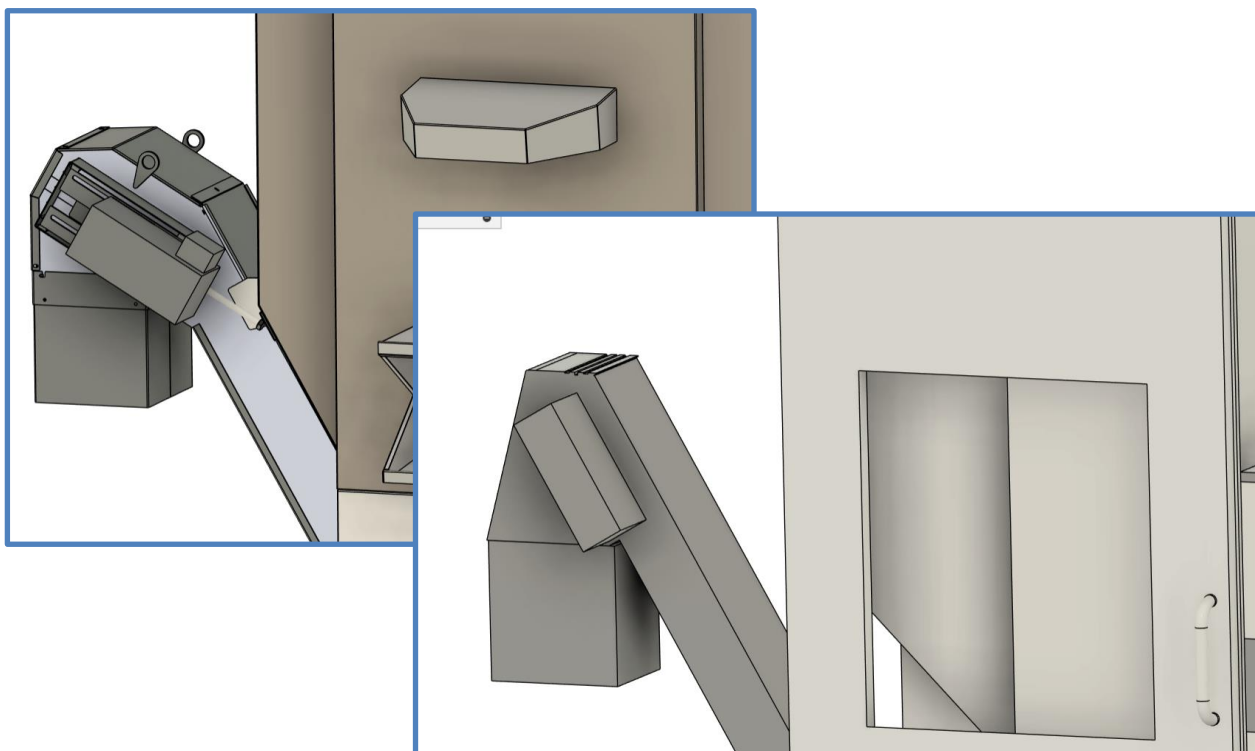


'Simplify' commands are now available – only after turning on the 'Capture Design History' described in the previous paragraph.

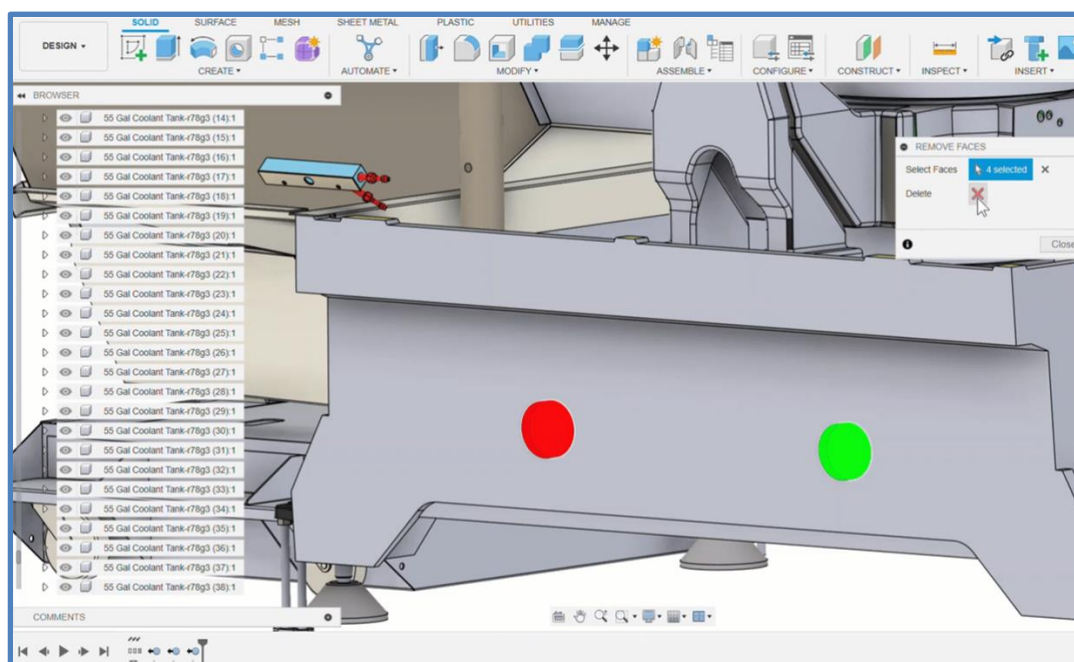
An example of something that should be removed is the clamp shown in the picture below. It is located outside the enclosure, serves no purpose during simulation, and it adds complexity to the model.



The pictures below show an example of the chip conveyor mechanism before and after simplification.

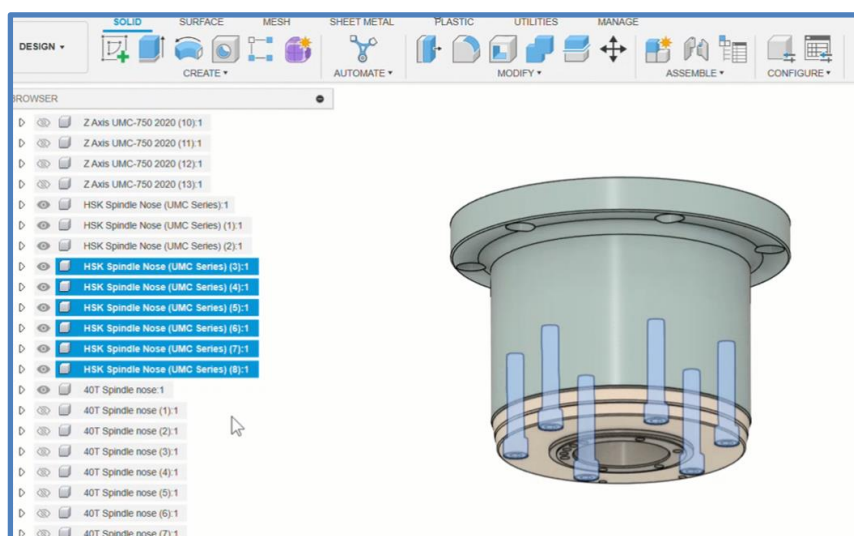


Remove unnecessary model parts (faces). The picture below is a simple example of removing a couple holes. The model should be made 'light weight' so Fusion can show, move, and collision check only the important bits.

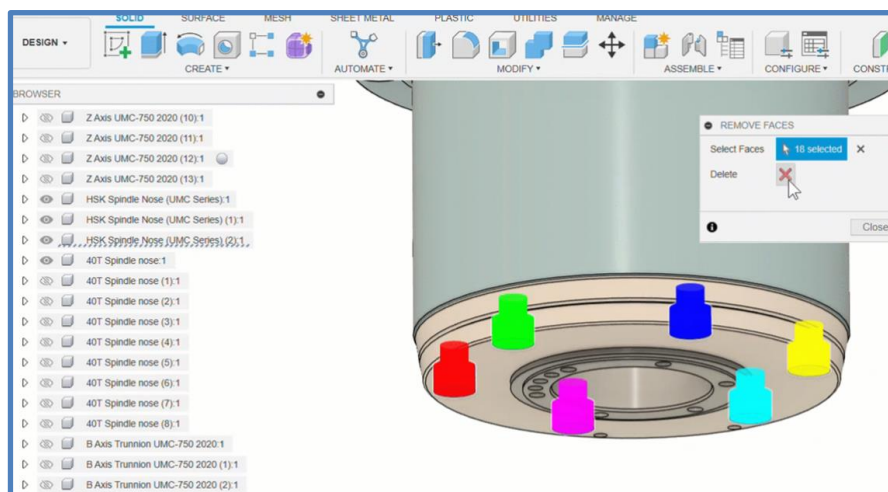


You might think of this process as like what back-packers do when preparing their gear for long hikes. Remove all unnecessary weight, even if it means cutting off half the toothbrush handle.

The model parts should be as simple as possible, but still be realistic so collision detection will be accurate. The picture below shows removing bolts that will have no effect on collision checking.

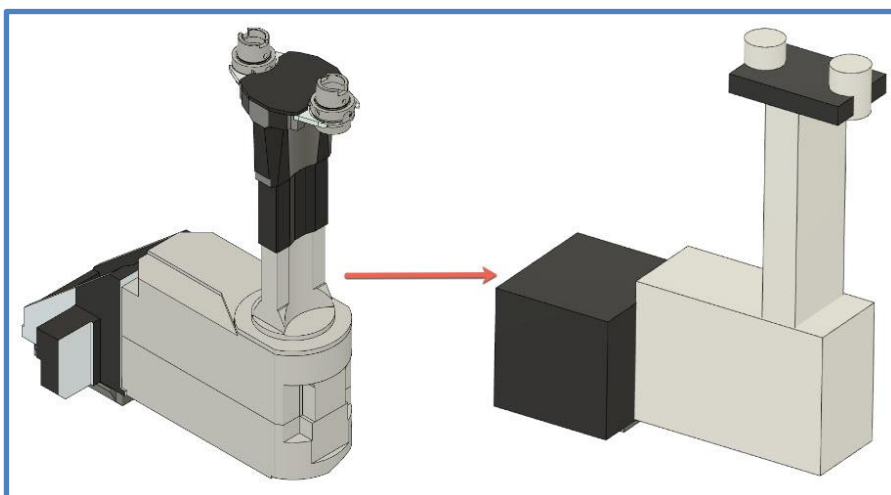


The picture below shows eliminating holes the bolts were in by using 'Simplify' and 'Remove Features'.



Generally inner cavities should be deleted. Some can be left for visual style, but for example, inside the spindle taper, higher up holes and chamfers are not necessary.

The part shown below may seem to be too simplified, but it's fundamental to obtain a well-defined but lightweight model. Strive to eliminate hidden parts, and in general most parts not affected by the collision check. This 'streamlining' of the model facilitates rapid simulation but does not interfere with accurate collision checking.



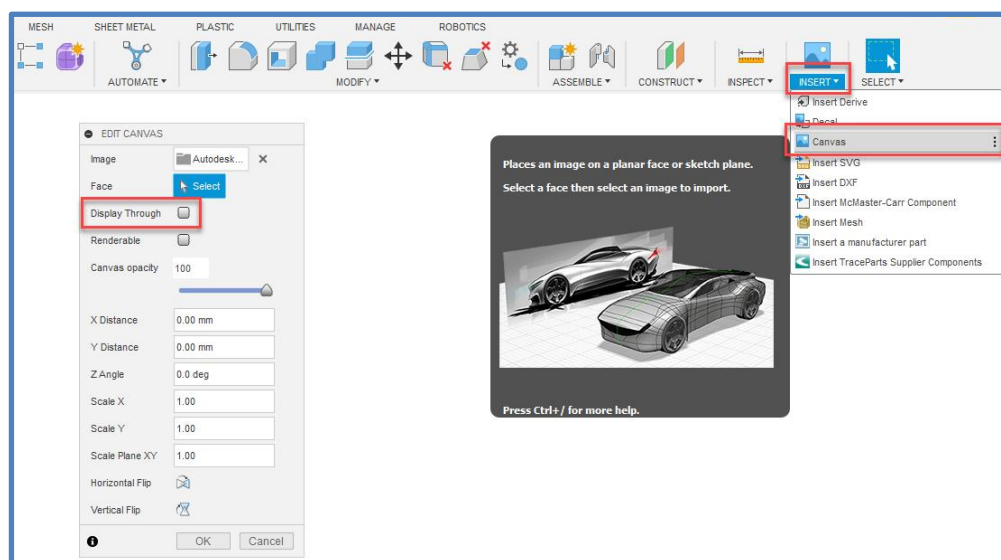
c. Add logo and stickers

Depending on what data you have, you can add the machine logo as a 3d part or a canvas.

If it is a 3D model, you should treat it like any other model component.

Do keep in mind that if your data came from the machine tool manufacturer and did not have company logos or model numbers, you should check with them to see if they want them included.

If you have the logo in image format, or want to add any other stickers, you can create a sketch plane of the correct size to which you can apply the image using the command 'insert-canvas'. However, it is recommended to use the 3D model for a better definition of the logo.

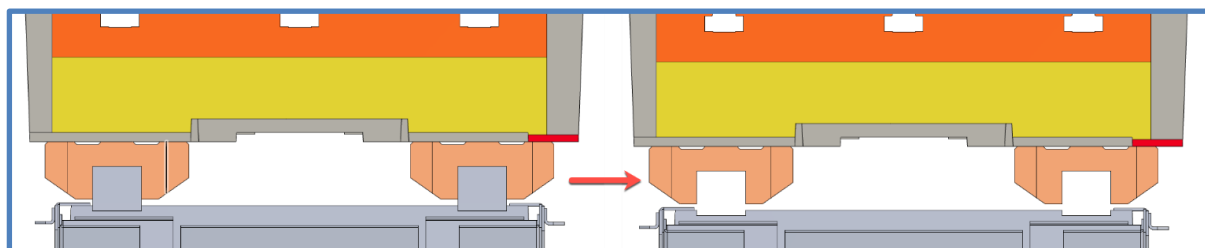


Key things to remember:

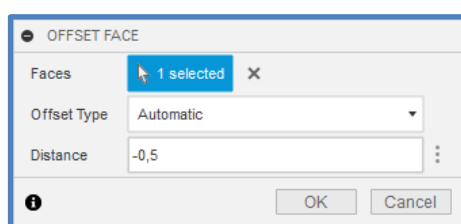
- In Canvas command, deactivate 'Display Through' option, as shown above.
- Whenever possible, it is better to use the solid rather than the image because the image generates an external profile that cannot be eliminated.

D. Colliding components

To reduce the possibility of collisions being detected on parts that are attached, there should be a gap created between each of the components. This usually means removing bearings & linear axis rails as shown below:

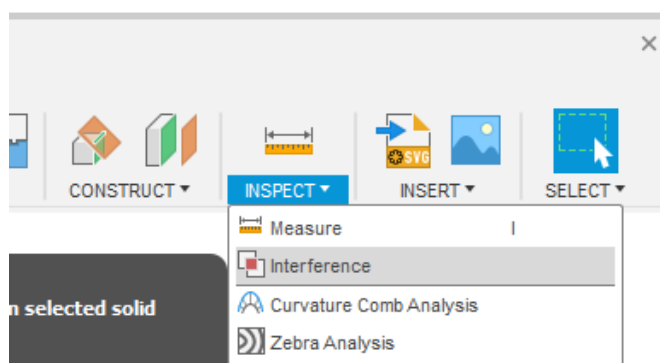


An alternate method is to create a small offset between the parts using 'Press-Pull' command shown in the picture below.

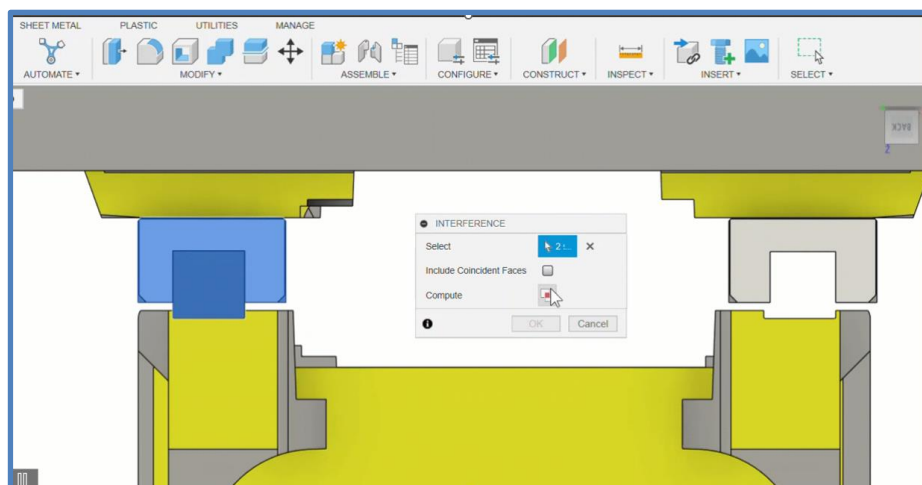


E. Interference Computation

The 'Interference' command found on the Inspection drop down is useful for detecting components too close together. Interference between components should be eliminated.



The picture below shows checking for interference between bearing parts. This is done by selecting the two components, then using the 'compute' button.



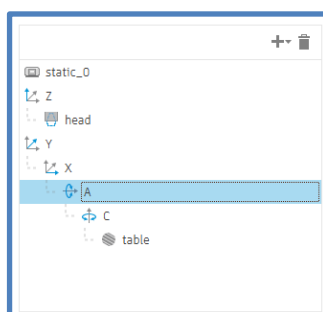
3. MAKE A MACHINE DEFINITION

Once the model preparation is complete, you need to create a Machine Definition. You can then add your machine model to the Machine Definition. (If you already have a Machine Definition you can use it rather than creating a new one.)

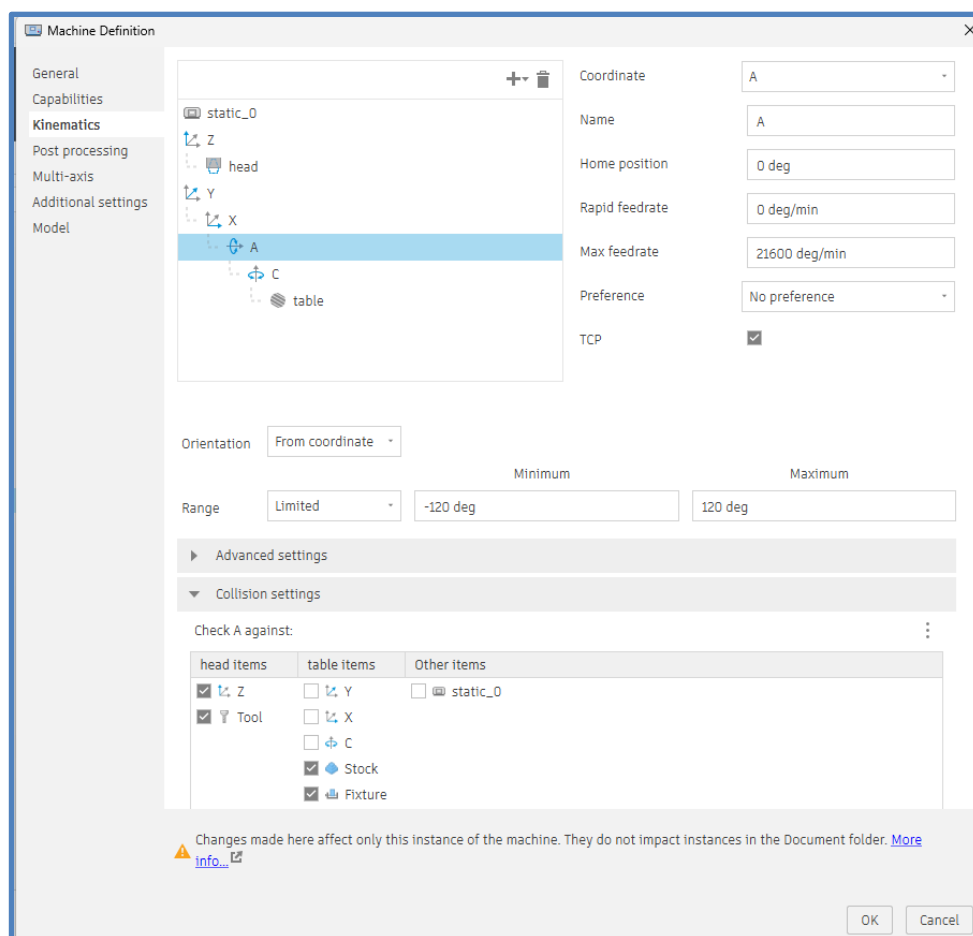
- 1) Change from the 'Design' workspace to the 'Manufacture' workspace.
- 2) Use the 'Machine Library' button to access the library.
- 3) Choose the menu item 'Local'.
- 4) Use the 'Plus sign' to create a new 'Milling' 'Machine Definition'. Or select one from those shown in the dialog and use the pencil icon to edit it.
- 5) Fill-in the fields for Vendor, Model, and Description because it will enable you to keep things organized and make it easier to come back here for edits.

Configuring the Kinematic tree:

- 1) The kinematic tree must represent the exact order of axes as they are in the machine.



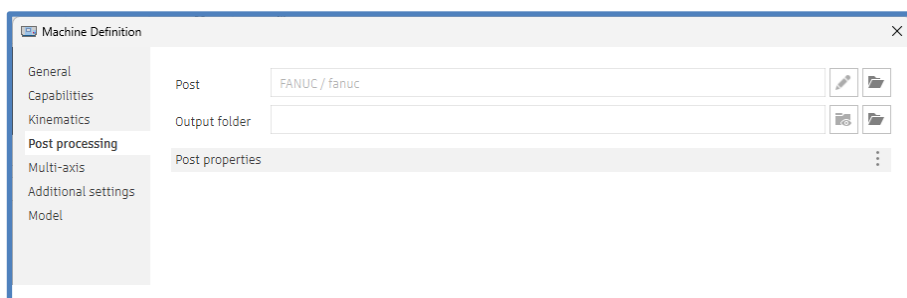
- 2) For multi-axis machines, the rotary axis information is crucial for sending data to the post processor to create NC programs. Here are some important settings to consider:
- TCP: Controls whether the tool center point axis is supported.
 - Orientation: Determines the rotation of the selected rotary axis and the axis it rotates around.
 - Range: Specifies the supported range of the selected axis.
 - Preference: Specifies the preferred direction of tilt/rotation of the axis.



- 3) The collision setting allows you to control axis collision checking against other axes. Note that enabling the collision setting for multiple items could slow down the verification process during simulation.

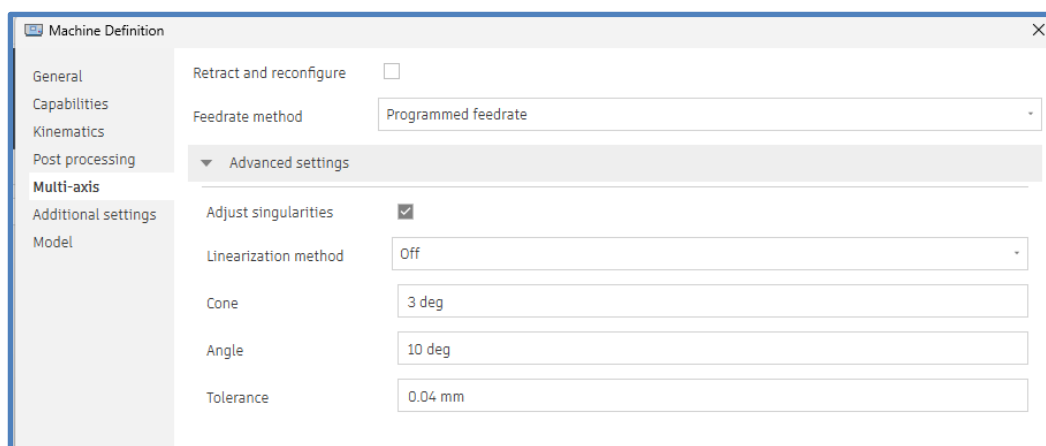
Post Processing:

This tab allows you to set the default post processor and output folder for the machine. By doing this, when the machine is selected in the NC program, it automatically sets the corresponding post and output folder.



Mutiaxis setting:

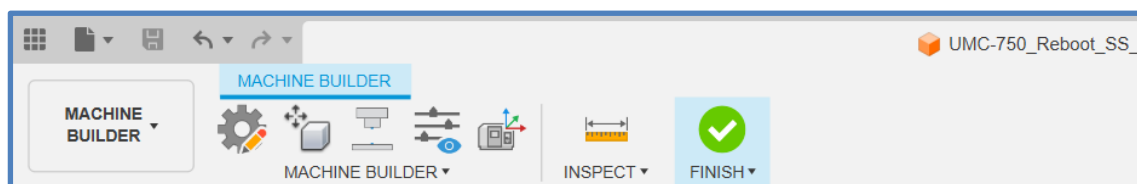
- 1) This setting allows you to control some of the multi-axis settings during post-processing:
 - a. Retract and Reconfigure: Enables the machine to rewind during simultaneous toolpaths. This occurs when the rotary axis reaches one of its limits, requiring you to reposition the tool. This mostly happens when both rotary axes are limited to a certain range.
 - b. Feedrate Method: Specifies the preferred feedrate method during post-processing of multi-axis toolpaths.
 - c. Adjust Singularities: Optimizes the simultaneous multi-axis toolpaths for singularities.



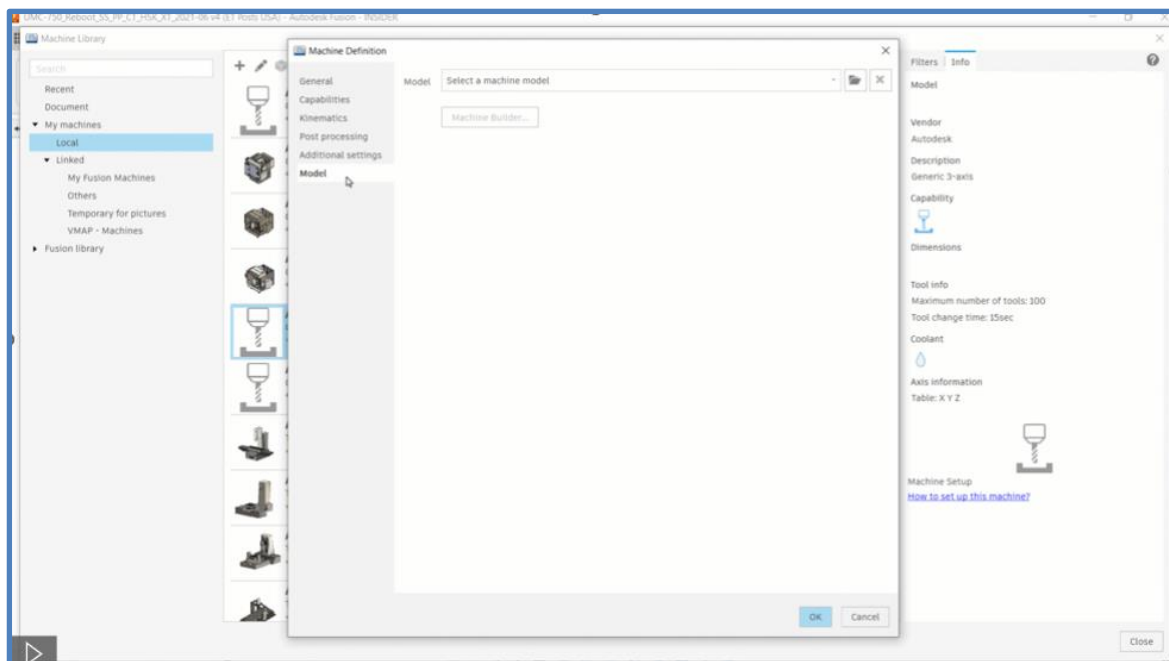
Save the machine definition and proceed to select the machine model for the next step.

4. ADD YOUR MACHINE MODEL TO A MACHINE DEFINITION

Once you have a Machine Definition created using the large 'plus sign' icon, or if you are using an existing Machine Definition that you are editing, follow these steps.



- 1) Click on the menu choice 'Model' (near the bottom of the menu on left).
- 2) The center section has a drop-down list titled 'Model'. Use it to select the item with 'Current document' in parentheses at the end.
- 3) Then directly under the drop-down list is the 'Machine Builder' button. Click on it to move into the 'Machine Builder' contextual environment. It is like a separate workspace with its own tool bar.
- 4) Start with the 'Setup Machine Model' icon.

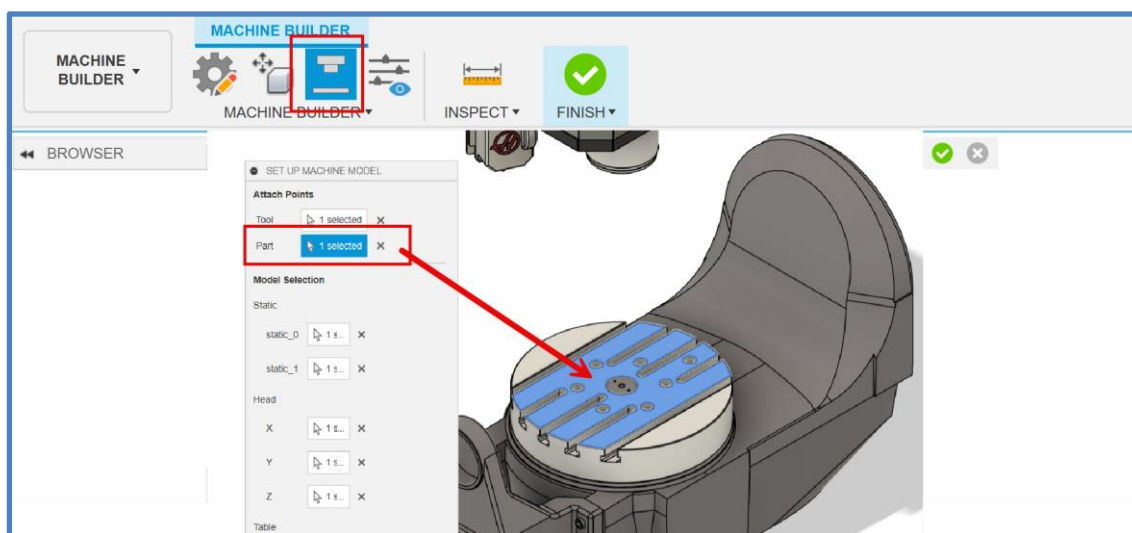


5. USE THE 'SET UP MACHINE MODEL' BUTTON TO LINK YOUR MODEL TO FUSION

The 'Setup Machine Model' button brings up a dialog box you use to effectively pair-up the model parts you have carefully named, to Fusion's record of what those parts are.

The tool tips should provide enough explanation to help you along in supplying the information for each of the fields. One thing to be careful of is to avoid accidentally selecting surfaces while defining the Tool/Part attachment and the rotation points.

There are links at the end of this document to videos that walk step by step through the process of connecting the model components to Fusion.



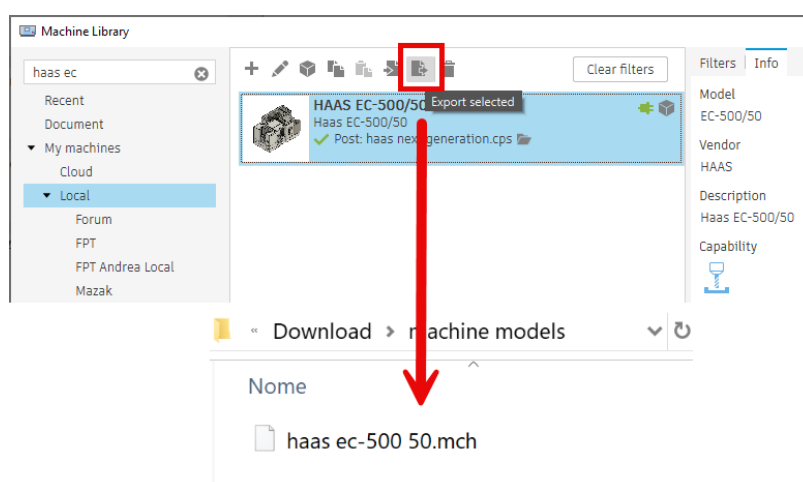
After you have paired-up the components in the model to the fields in the 'Setup Machine Model' dialog box, use the icon for previewing machine movements (icon looks like 3 sliders) to test that you have the correct settings. It is strongly recommended that you test these motions before continuing.

When you are satisfied that you have the machine model parts connected to Fusion correctly and tested the motions using 'Preview Machine Movement', use the icon with large check mark on a green background to return to the Manufacturing workspace.

If you then discover changes are needed, you can get back to the Machine Builder workspace by going to the Machine Library, selecting the machine from the list in the center of the dialog, then using the pencil icon (just right of the large plus icon) to reenter the Machine Definition where you can use the menu choice Model again.

6. EXPORT MACHINE DEFINITIONS AND MODELS

If you want to send a Machine Definition to another Fusion user, you can simply export the machine from the library as shown in the picture below.



If you want to send an entire machine model to another user, you can export the Fusion model by using the File pull down and selecting 'Export...'.

And you can do the same export from the Machine Library dialog.

7. USER FEEDBACK

If you would like to provide us with feedback on this document, or suggestions for other documents like this, please send email to:

channel.posts@autodesk.com

Please set the subject of the email to read: Feedback on Milling Machine Subtractive

8. USEFUL LINKS

The links below provide additional information and instructions on related subjects.

- [Fusion Help | Machines for manufacturing | Autodesk](#)
- [Machine Builder](#)
- [Workflow: Building a machine for machine simulation](#)
- [Fusion Help | Preview a machine's movements in the Machine Builder | Autodesk](#)